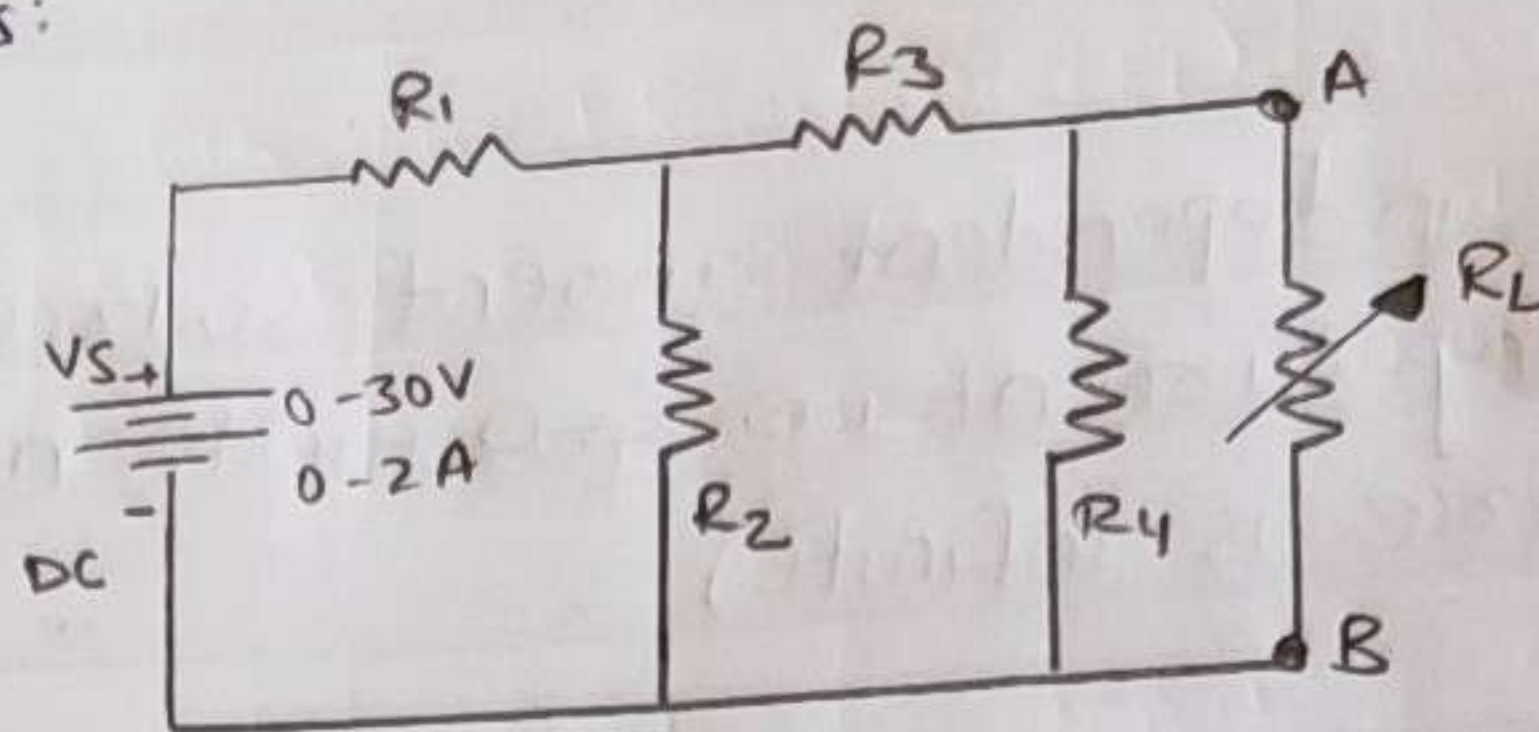
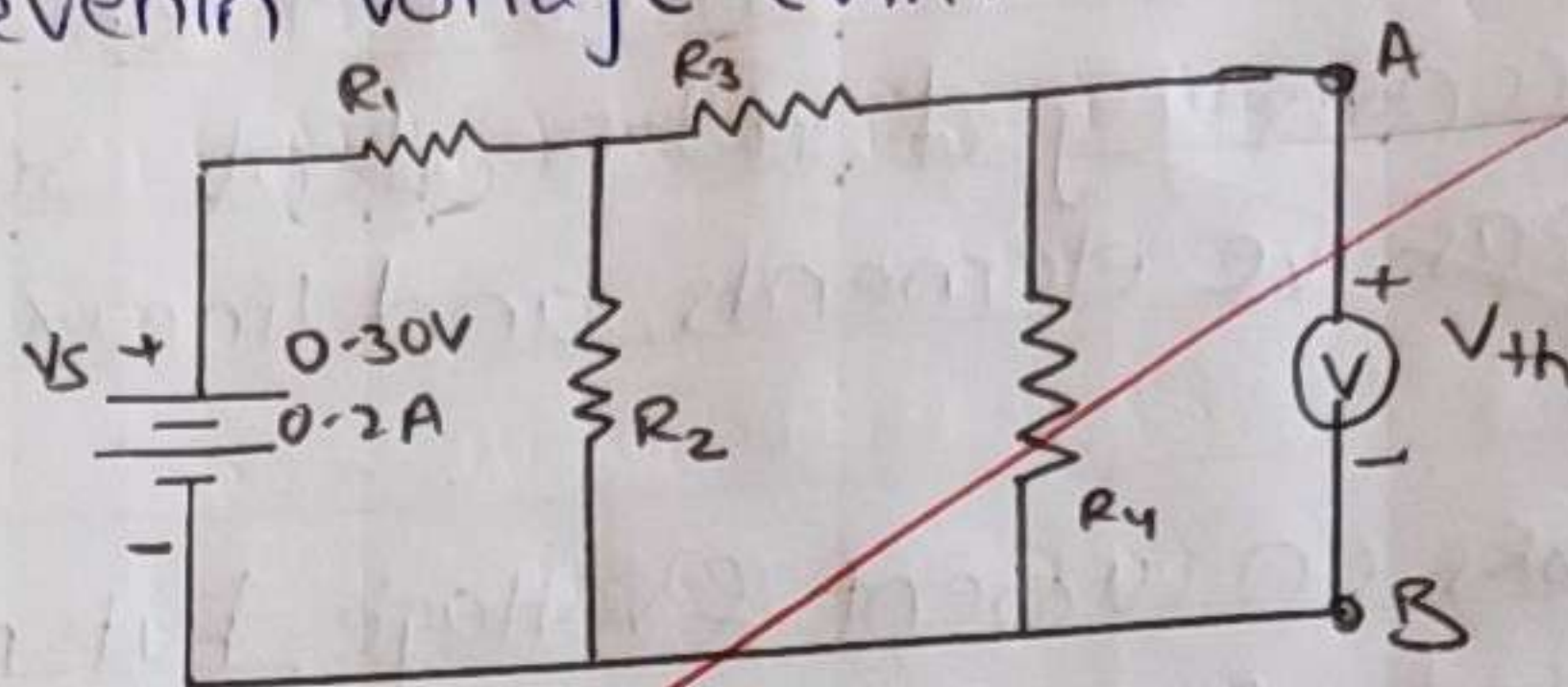


o Circuit Diagrams:

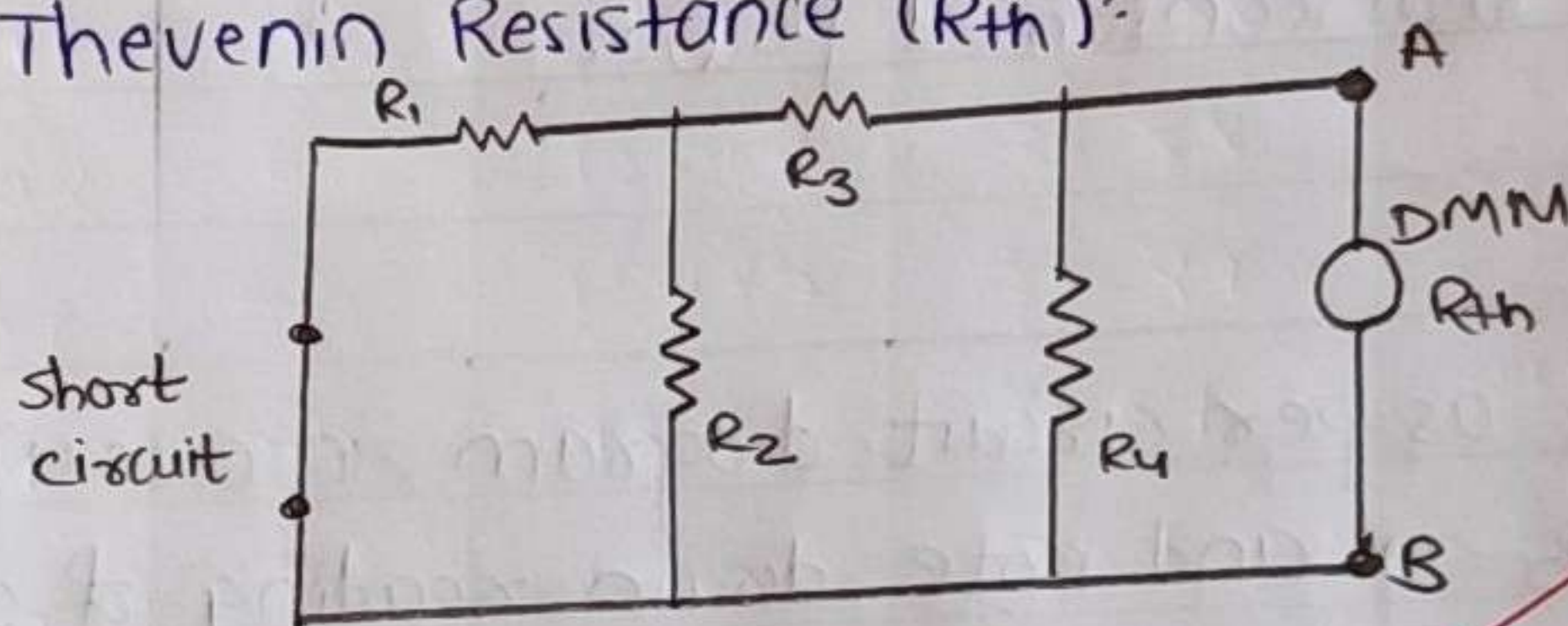
(i) Main Circuit:



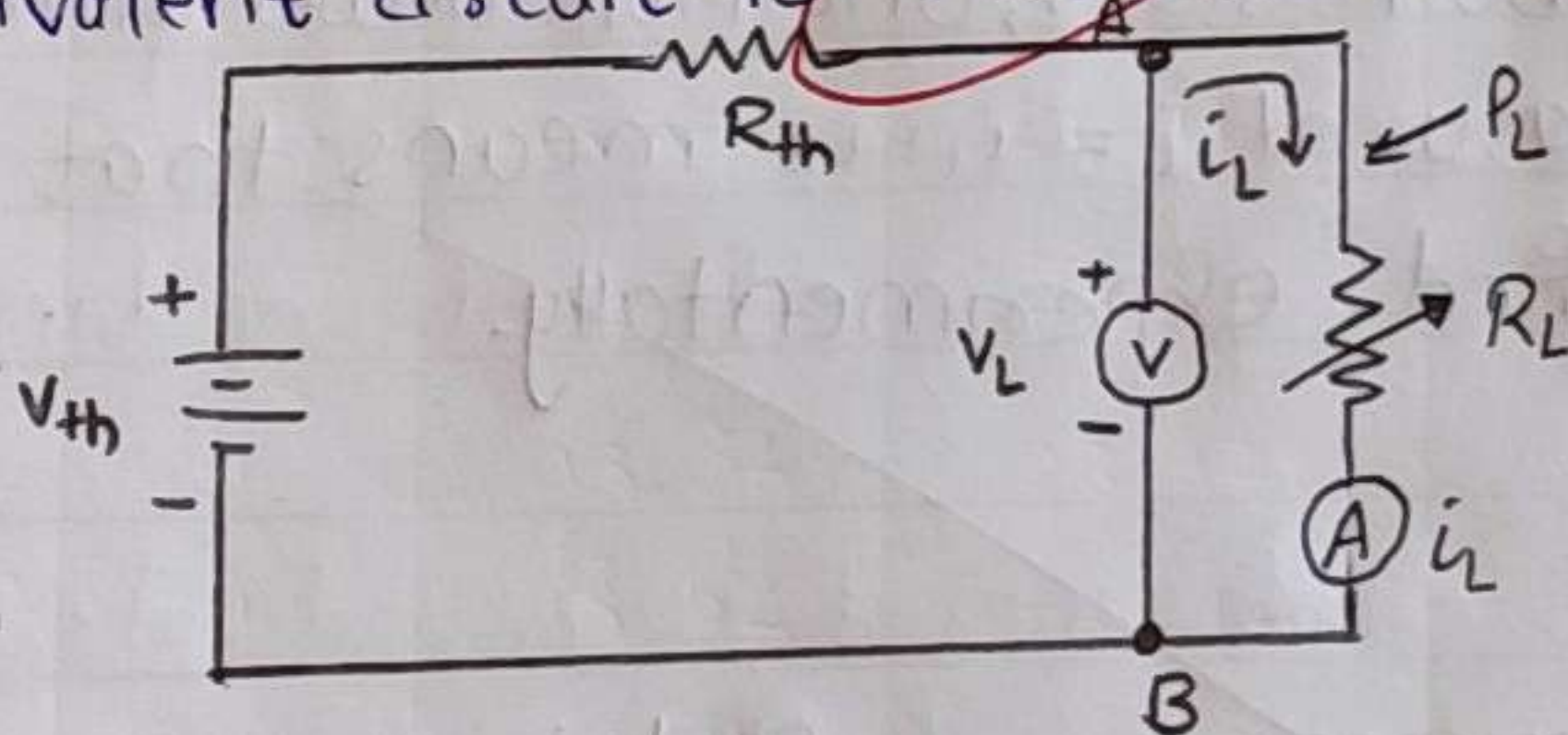
(ii) To measure Thevenin voltage (V_{th}):



(iii) To measure Thevenin Resistance (R_{th}):



(iv) Thevenin's Equivalent circuit to measure load power.



EXPT.
NO. 5NAME Thevenin's Theorem and Max. Power
Transfer Theorem (MPTT).

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YOUVA

Date: 03/10/24

◦ Aim:

- Experimental verification of maximum power transfer theorem with DC excitation.

◦ Apparatus:

Equipment	Range	Quantity
Multiple Power Supply	0-30V	1
Breadboard	—	1
Decade Resistance Box (DRB)	—	1
Resistors	1K Ω	3
Ammeter	0-200mA	1
Digital Multimeter (DMM)	—	1
Connecting wires	—	Required

◦ Theory:

- Thevenin's Theorem: Any linear two-terminal network can be replaced by an equivalent network consisting of a voltage source (V_{th}) in series with a resistance (R_{th}).

where V_{th} = open circuit voltage of two terminals

R_{th} = equivalent resistance at load terminal when the sources are made inoperative.

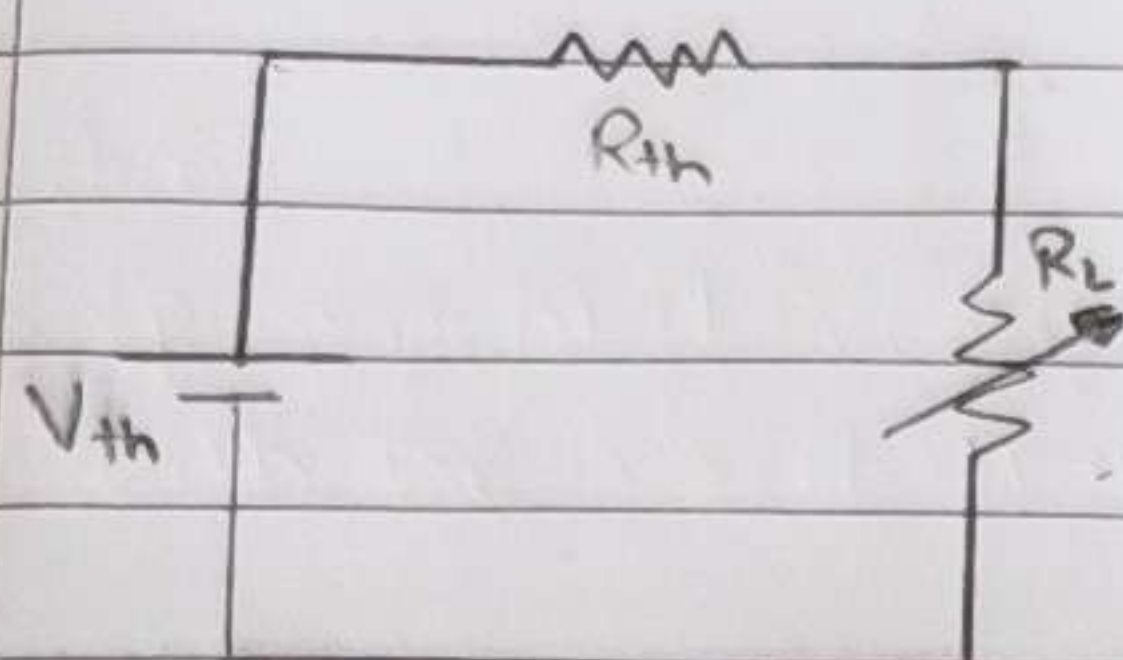
- Max. Power Transfer Theorem: A load will receive maximum power from a network when its resistance is exactly equal to the Thevenin's Resistance of the Network.

Teacher's Signature: _____

o observation Tables:

	V_{th}	R_{th}	Load Resistance (R_L)	P_{max}	Efficiency (η)
				0.015W	50%
Theoretical	6V	600 Ω	600 Ω	0.01515W	49.61%
Practical	5.98V	590 Ω	590 Ω		

Load Resistance $R_L(\Omega)$	Load Voltage $V_L(V)$	Load Current $i_L(mA)$	Load Power $P_L(W) = V_L \cdot i_L$	Input Power $P_{in}(W) = V_{th} \cdot i_L$	% efficiency $\eta = \frac{P_L}{P_{in}} \times 100$
150	1.2	7.5	9	44.85	20.06
200	1.45	7.5	10.875	44.85	26.75
250	1.6	6	9.6	35.88	29.09
300	1.74	6.5	11.31	38.87	32.27
350	1.9	6	11.4	35.88	34.70
400	2.08	6	12.48	35.88	37.90
450	2.27	5.5	12.485	32.89	39.70
500	2.4	5.5	13.2	32.89	41.8
550	2.5	5	12.5	29.9	46.8
600	2.8	5	14	29.9	49.6
650	2.97	4.5	13.365	26.91	51.17
700	3.07	4.5	13.815	26.91	52.9
800	3.23	4	12.92	23.92	54
850	3.33	4	13.32	23.92	55.60
900	3.38	4	13.52	23.92	56.50
950	3.46	3.5	12.11	20.92	57.8
1000	3.5	3.5	12.25	20.92	58.5



We have, $i = \frac{V_{th}}{R_{th} + R_L}$

$$\therefore \text{Power to load, } P_L = i^2 R_L = \left(\frac{V_{th}}{R_{th} + R_L} \right)^2 R_L$$

For maximum power, $P_L' = 0$.

$$\therefore \frac{\partial P_L}{\partial R_L} = V_{th}^2 \left[\frac{(R_L + R_{th})^2 - 2R_L(R_L + R_{th})}{(R_{th} + R_L)^4} \right] = 0$$

$$\therefore (R_L + R_{th})^2 = 2R_L(R_L + R_{th})$$

$$\therefore R_L + R_{th} = 2R_L \Rightarrow \boxed{R_L = R_{th}}$$

$$\therefore P_L = \left(\frac{V_{th}}{R_{th} + R_{th}} \right)^2 R_{th} = \frac{V_{th}^2}{4R_{th}}$$

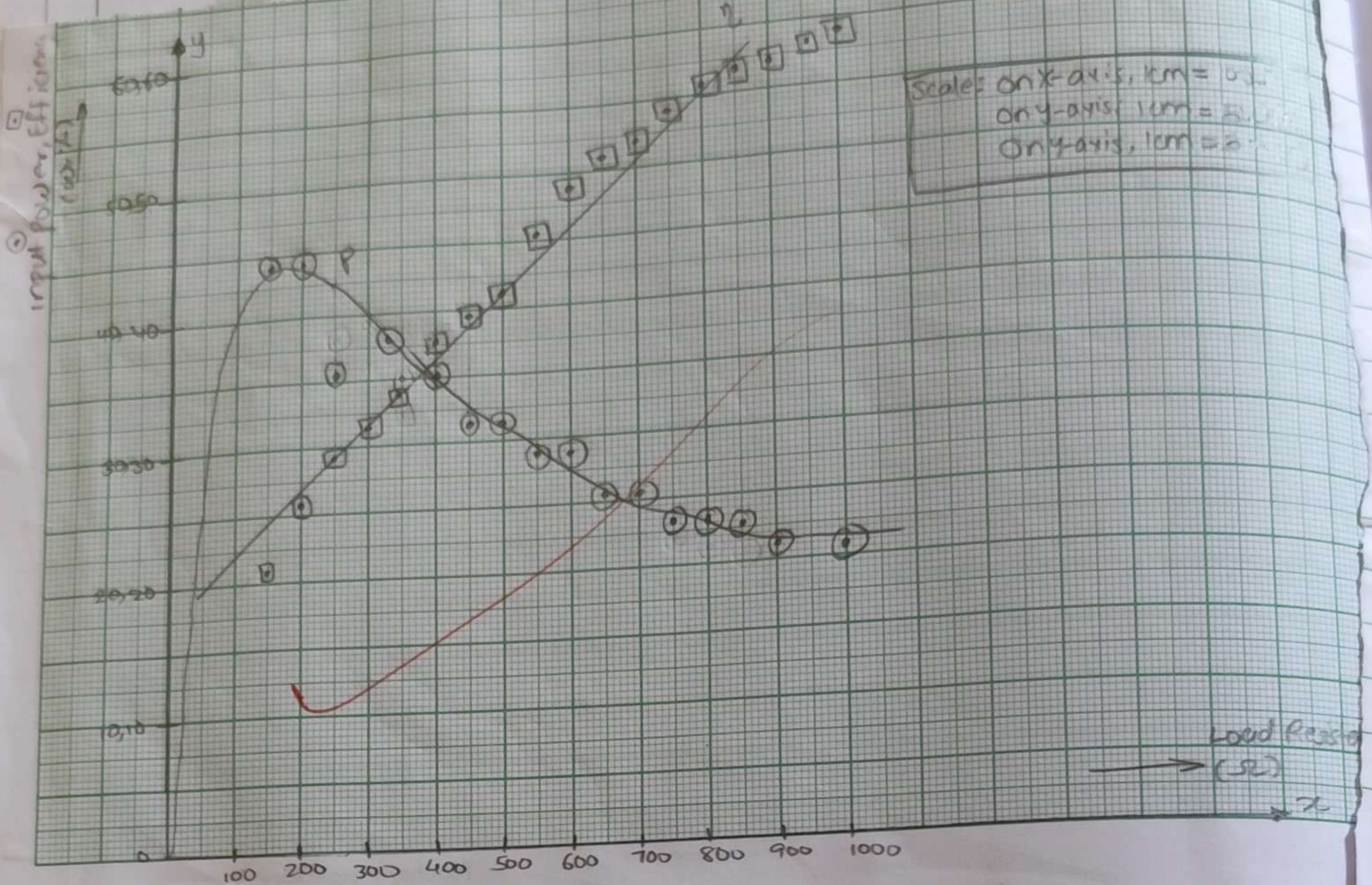
$$\text{Input power, } P_{in} = V_{th} \cdot i_L = V_{th} \left(\frac{V_{th}}{R_{th} + R_{th}} \right) = \frac{V_{th}^2}{2R_{th}}$$

$$\therefore \text{Efficiency } = \eta = \frac{P_L}{P_{in}} = \frac{V_{th}^2/4R_{th}}{V_{th}^2/2R_{th}} = 0.5 = 50\%$$

Procedure:

- To find Thevenin voltage (V_{th}), apply voltage $V_s = 10V$ at source terminals and remove load resistance R_L . The voltage across the open circuited terminals is noted.
- Short circuit the voltage source in the given circuit after removing the load resistance. Thevenin's Resistance (R_{th}) is measured with the help of digital multimeter (DMM).
- Construct the Thevenin Equivalent circuit.

graph:



- By varying the load resistance value using Decade Resistance Box, we tabulate the three values load voltage, current, and power.
 - Plot the graphs Load Resistance (R_L) vs Load Power (P_L) and the % Efficiency (η).
 - Identify the maximum power point and efficiency.
- o Results:
- Thevenin's Theorem and Max. Power Transfer Theorem (MPTT) have been proven experimentally. At max-power, Thevenin's Resistance and Load Resistance are found to be equal.

~~Imp
P.M.~~